PROJECT EXECUTABLE FILES

Comic Craft – AI Comic Story Creator using Gemini Models

Prepared from the supplied Project Executable Files template and Comic Craft project documentation

Field	Details
Date	15 March 2026
Team ID	SWTID-2026-8237
Project Name	Comic Craft – AI Comic Story Creator using Gemini Models
Maximum Marks	3 Marks

Project Executable Files

This document presents the executable-file submission information for ComicCraft. The supplied project describes a web-based AI comic creator using FastAPI, Google Gemini models, Stable Diffusion and FPDF. The application accepts story preferences, generates a five-panel comic, previews it in the browser and exports the result as a PDF.

Step 1: Submission Checklist

S.No	Item to Submit	Status	Project-specific details
1	Complete source code (all files and folders)	Yes / documented	Application modules include routes.py, gemini_flash.py, gemini_pro.py, image_generator.py, layout_builder.py and exporters.py.
2	README / Setup Guide	Yes / documented	Local setup uses a Python virtual environment, dependency installation, environment variables and Uvicorn.
3	requirements.txt / package.json / dependency file	Yes / documented	Python dependencies are documented; the project uses FastAPI, Uvicorn, Jinja2, Gemini, Diffusers, FPDF, Pillow and related packages.
4	Database schema / seed files (if applicable)	NA	No user database is documented; generated

S.No	Item to Submit	Status	Project-specific details
			images/PDFs are handled through project static folders.
5	Environment configuration file (.env.example or similar)	Yes / required	GEMINI_API_KEY and HF_API_KEY are configured through environment variables.
6	Deployed application URL (if hosted)	Not provided	The supplied material documents local deployment; no hosted URL is supplied.
7	APK / Executable binary (if applicable)	NA	ComicCraft is documented as a web application, not a mobile/desktop binary.
8	Dockerfile / Containerization config (if applicable)	Not provided	No Docker configuration is specified in the supplied project material.
9	Test files and test results	Partial / documented	The project provides API/image testing routes, but separate automated test files/results are not supplied in the source material.
10	Demo video or walkthrough (if required)	Not provided	No demo-video link is supplied in the project material.

Step 2: File / Folder Structure

The following structure is based on the modules, templates, static assets and routes described in the supplied ComicCraft documentation:

ComicCraft/

├── app/

| ├── main.py

| ├── routes.py

| ├── gemini_flash.py

| ├— gemini_pro.py

| ├— image_generator.py

| ├— layout_builder.py

| └— exporters.py

├— templates/

| ├— index.html

| ├— comic_preview.html

| └— export_success.html

├— static/

| ├— panels/

| └— exports/

├— requirements.txt

└— .env (credentials/configuration)

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://github.com/Muthu sundar /comic Craft-AI
Login	https://myskillwallet.ai/login
Credentials (Demo)	https://myskillwallet.ai/login
Platform / Hosting Provider	https://www.skills.google/public_profiles/0628f4e4-51bf-4530-b9f9-df536d95e4f0
Repository Link	https://github.com/Muthu sundar/comicCraft-AI
Demo Video Link	https://drive.google.com/file/d/1wcvhZo-ivVfnvekELCtnJXY9bXqorjPY/view?usp=drivesdk

Comic Craft Generation Workflow

Step	Process	Technology / Component
1	Collect story prompt, character name, setting, tone and art style	HTML/CSS/Jinja2 + FastAPI
2	Generate a structured five-panel outline	Gemini Flash

Step	Process	Technology / Component
3	Generate narration and character dialogue	Gemini Pro
4	Generate comic-style panel illustrations	Stable Diffusion via Hugging Face Diffusers
5	Build the organized comic panel layout	layout_builder.py
6	Preview the generated comic in the browser	Jinja2 templates
7	Export the complete comic as a downloadable PDF	FPDF / exporters.py

Step 4: Run Instructions

1. Install Python and ensure the required project dependencies are available.

2. Create and activate a virtual environment, for example: python -m venv comiccraft-env.

3. Install the documented dependencies using requirements.txt or the project dependency list.

4. Configure GEMINI_API_KEY and HF_API_KEY as environment variables; do not hard-code credentials.

5. Run the application with: uvicorn app.main:app --reload.

6. Open the local application at http://127.0.0.1:8000.

7. Use /docs at http://127.0.0.1:8000/docs for FastAPI API documentation/testing.

8. Enter the comic prompt and preferences, generate the comic, review the preview and export/download the PDF.

Documented Routes / Access Points

Route	Purpose
/	ComicCraft web interface/home page
/generate	Form-based comic generation flow
/generate-comic/json	JSON-based comic generation API
/test-image	Direct image-generation testing route
/export-success	PDF export confirmation page
/docs	FastAPI interactive API documentation

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	No hosted/deployed URL is supplied.	Use the documented local Uvicorn deployment.
2	Gemini and Hugging Face credentials are required.	Set GEMINI_API_KEY and HF_API_KEY through environment configuration.
3	No user authentication/account management is included in the current version.	The documented workflow is intended for single-comic generation.
4	No specific production response-time or concurrency targets are documented.	Current documentation focuses on successful local execution.
5	No Dockerfile or containerization configuration is supplied.	Run using the documented Python virtual environment and Uvicorn workflow.
6	Repository and demo-video links are not supplied.	These fields should be added if available before final submission.
7	Separate automated test files/results are not shown in the supplied material.	Use the documented /docs and /test-image routes for functional checking.

Functional Access Summary

The supplied project documentation identifies eight implemented functional features: user input and customization; AI comic outline generation; story and dialogue generation; comic illustration generation; comic panel layout; comic preview; PDF export/download; and API/image testing.

Feature	Status
User Input & Customization	Done
AI Comic Outline Generation	Done
Story & Dialogue Generation	Done
Comic Illustration Generation	Done
Comic Panel Layout	Done
Comic Preview	Done
PDF Export & Download	Done
API & Image Testing	Done

Conclusion

ComicCraft provides an end-to-end documented workflow from user prompt to downloadable comic. The executable submission should include the application source, setup/dependency information and environment configuration required for local execution. Fields not supported by the supplied project material—such as a hosted URL, repository link, demo video, Docker configuration or separate automated test results—are explicitly marked as not provided rather than being invented.